Architectural-items_detection: mini report

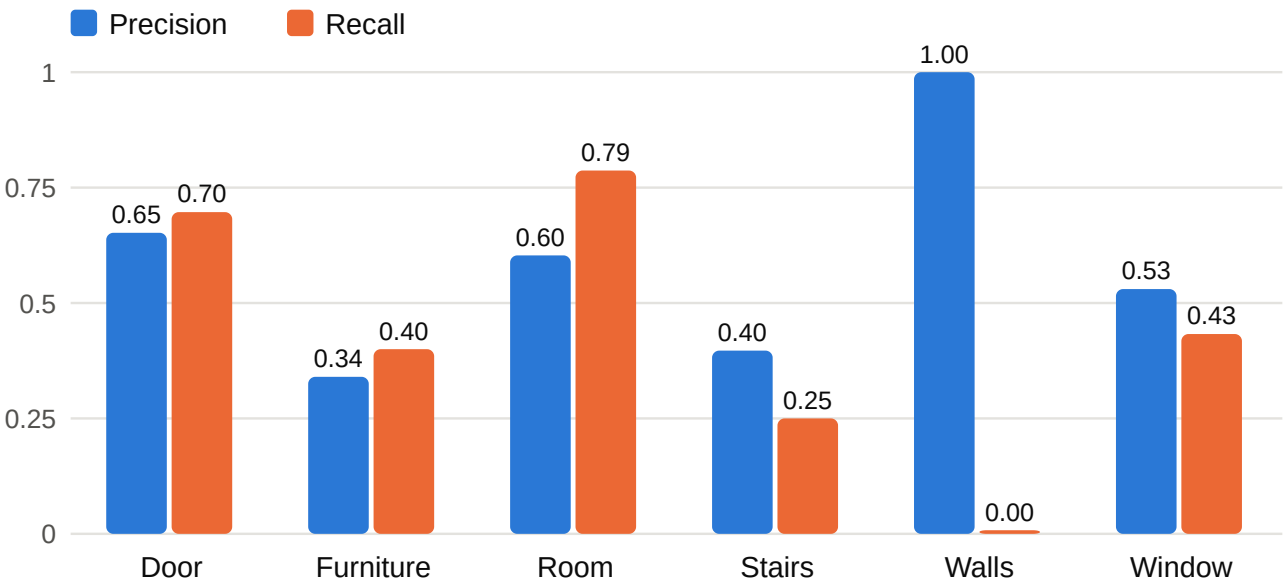
Cesare Della Corte · M4U3 · github.com/cesaredellacorte-bot/Architectural-items_detection

Executive summary

This project trains a YOLOv8n object detector to recognise six architectural elements (Door, Furniture, Room, Stairs, Walls, Window) in plan views, as a first step towards a fast, automatic recognition workflow for AECO. The data are a subset of about 110 images from the public Roboflow Universe dataset *cad-drawings* (MIT licence), which the author re-annotated with the six classes and split about 80/20. After 50 epochs of fine-tuning (batch 16, image size 640, Ultralytics 8.4.163, Google Colab) the model reaches **mAP50 0.40** and **mAP50-95 0.25** (precision 0.59, recall 0.43) on the validation split. Doors and rooms are detected reliably; windows, furniture, stairs and walls are not yet. The model is a first-pass assistant, not a measuring tool.

Results

Class	Precision	Recall	mAP50	mAP50-95
ALL	**0.587**	**0.428**	**0.396**	**0.247**
Door	0.652	0.697	0.723	0.449
Furniture	0.340	0.400	0.276	0.170
Room	0.603	0.787	0.740	0.526
Stairs	0.397	0.250	0.210	0.133
Walls	1.000	0.000	0.003	0.002
Window	0.530	0.433	0.425	0.202



Walls: precision 1.00 only reflects almost no predictions (recall 0.00).

Interpretation

- **What works:** Room (mAP50 0.74, recall 0.79) and Door (0.72, recall 0.70) meet the target of mAP50 and recall of at least 0.60. They are large or visually distinctive.
- **What does not:** Window (mAP50 0.43, recall 0.43) misses the target. Furniture has the lowest precision (0.34): its symbols are ambiguous. Stairs (40 examples) and Walls (3 examples) have too little data; Walls precision of 1.0 only means almost no predictions were made.
- **Precision vs recall:** recall (0.43) is lower than precision (0.59), so the model misses more objects than it invents. mAP50-95 (0.25) is much lower than mAP50 (0.40), so boxes are often loosely placed.
- **Data, not model size:** in the final run mAP50 is still rising slowly at epoch 50 while precision is flat at about 0.6, and the validation set is small (a few dozen images), so figures are noisy.

Error analysis (hypotheses)

False positives: extra Furniture boxes where furniture may be unlabelled or ambiguous; Window boxes on wall lines and dimension marks; duplicate nested Room boxes. **False negatives:** Stairs (rare, varied styles); Walls (long thin shapes, 3 examples); Windows in low-contrast scans and hand-drawn plans. Image-level examples are in the repository (*results/evidence/errors/*, *docs/error_analysis.md*).

Limitations

- About 110 images of limited style, single annotator, no inter-annotator check; not validated on other countries, scales or drawing standards.
- Do not use the output for quantities, permits, compliance or safety decisions without human verification. A missed door or stair is worse than a false alarm, so a human should review every plan.
- Licences: images MIT (cad-drawings, cited in the README); code and annotations MIT; weights derive from Ultralytics (AGPL-3.0).

Next steps (three data improvements)

1. Audit labels: write explicit Furniture and Room rules, add missing boxes. Target: Furniture precision from 0.34 to above 0.50.
2. Add 40-60 stairs and hard window examples (scans, hand-drawn) plus 10-15 background images. Target: Stairs recall from 0.25 to above 0.50, Window recall from 0.43 to above 0.60.
3. Enlarge the validation set to 50+ plans split by source plan; redefine or drop Walls.

Reproducibility

Open *notebooks/02_train_eval.ipynb* from the repository in Google Colab, select a GPU, Restart session, Run all. Metrics, curves and evidence are written to *results/*.